

Design Process

4

4.1 Design Rule : Principles that Support Usability

Q.1 What is design rule ?

Ans. : Design rule

- Design rules (or usability rules) are rules that a designer can follow in order to. Increase the usability of the system/product e.g., principles, standards, guidelines.
- Designing for maximum usability is the goal of interactive systems design.
- Abstract principles offer a way of understanding usability in a more general sense, especially if we can express them within some coherent catalog.
- Design rules in the form of standards and guidelines provide direction for design, in both general and more concrete terms, in order to enhance the interactive properties of the system.
- The essential characteristics of good design are often summarised through 'golden rules' or heuristics.
- Design patterns provide a potentially generative approach to capturing and reusing design knowledge.

Q.2 What is learnability, flexibility and robustness in context of usability ? It has been suggested that consistency could be considered a major category of interactive principles, on the same level as learnability, flexibility and robustness. If this had been the case, discuss the principles that would appear in support of consistency ?

 [SPPU : Dec.-17, End sem, Marks 8]

(4 - 1)

Ans. :

- The consistency, it can take many forms, because it is usually referred to in relation to some other feature of the interaction between user and system. There are consistency related to the following principles :
- Familiarity - consistency with respect to prior real-world experience.
- Generalizability - consistency with respect to experience with the same system or set of applications on the same platform.
- In addition, we could interpret some other principles as contributors to consistency :
- Affordance - consistency with understood intrinsic properties of an object, so a soft button on the screen should allow us to always 'push' on it to select some action :
- Predictability - consistency of system response with user's expectation, given the user has some information about past interaction history.
- Substitutivity - consistent permission from system to allow use of equivalent values for input and output.
- Commensurate effort - consistency of effort with respect to doing and undoing tasks.
- Response time stability - consistency of system response for similar actions.
- Some other principles for consistency from the text and elsewhere :
- Consistency can be relative to the form of input/output expressions relative to the user's conceptual model of the system. An example in the text involves using keys whose relative positions are similar to commands for the systems (any set of four typewriter keys that form a diagonal to indicate up, down, left and right information for an input command).
- Consistency can be with respect to social or cultural conventions (e.g., using red to indicate stop or hot, green for go, blue for cool).

Q.3 Define the following : Learnability, Predictability, Generalizability and Consistency.

Ans. :

- Learnability concerns the features of the interactive system that allow novice users to understand how to use it initially and then how to attain a maximal level of performance.
- Predictability is a user-centered concept; it is deterministic behavior from the perspective of the user. It is not enough for the behavior of the computer system to be determined completely from its state, as the user must be able to take advantage of the determinism.
- The generalizability of an interactive system supports this activity, leading to a more complete predictive model of the system for the user. We can apply generalization to situations in which the user wants to apply knowledge that helps achieve one particular goal to another situation where the goal is in some way similar. Generalizability can be seen as a form of consistency.
- Consistency relates to the likeness in behavior arising from similar situations or similar task objectives. Consistency is probably the most widely mentioned principle in the literature on user interface design.

Q.4 What is task migratability ?

Ans. :

- It is transfer of control for execution of tasks between system and user. It should be possible for the user or system to pass the control of a task over to the other or promote the task from a completely internalized one to a shared and cooperative venture.
- Hence, a task that is internal to one can become internal to the other or shared between the two partners. Example of task migratability is spell checking. It is equipped with a dictionary, you are perfectly able to check your spelling by reading through the entire paper and correcting mistakes as you spot them.

- This task is perfectly suited to automation, as the computer can check words against its own list of acceptable spellings.
- It is not desirable, to leave this task completely to the discretion of the computer, as most computerized dictionaries do not handle proper names correctly, nor can they distinguish between correct and unintentional duplications of words.
- In those cases, the task is handed over to the user. The spell-check is best performed in such a cooperative way.
- In safety-critical applications, task migratability can decrease the likelihood of an accident. For example, on the flight deck of an aircraft, there are so many control tasks that must be performed that a pilot would be overwhelmed if he had to perform them all.

Q.5 Explain the following terms

- Predictability
- Synthesizability
- Familiarity
- Consistency

[SPPU : Dec.-18, End sem, Marks 8]

Ans. :

- Predictability** : Support for the user to determine the effect of future action based on past interaction history. Predictability of an interactive system means that the user's knowledge of the interaction history is sufficient to determine the result of his future interaction with it.
- Synthesizability** : Support for the user to assess the effect of past operations on the current state.
- Familiarity** : The extent to which a user's knowledge and experience in other real-world or computer based domains can be applied when interacting with a new system. Familiarity has to do with a user's first impression of the system. Effective use of the affordances that exist for interface objects can enhance the familiarity of the interactive system.

iv) **Consistency** : Likeness in input-output behavior arising from similar situations or similar task objectives. Consistency relates to the likeness in behavior arising from similar situations or similar task objectives. Consistency is probably the most widely mentioned principle in the literature on user interface design.

Q.6 Describe any four usability goals of internet explorer.

[SPPU : May-19, End sem, Marks 8]

Ans. :

- Goals are needed that help to ensure usability :
 1. Ascertain users' needs
 2. Ensure proper reliability
 3. Promote appropriate standardization, integration, consistency and portability
 4. Complete projects on schedule and within budget
- Offer a graphical interface to the document, controlled by the mouse. Hypertext links are shown by highlighting the text that acts as the link in an alternative colour and are activated by clicking on the link.
- Allows users to view and navigate web pages on the Internet. Internet Explorer is the most widely used browser in the world.
- Nonstandard behaviours include support for vertical text, support for a variety of image effects and page transitions and support for embedding EOT fonts in webpages. The favicon feature allows web pages to specify a 16 × 16 pixel image for use in bookmarks.
- The browser contains most of the functionality required to view a web document, supporting text and graphics in an integrated package, special file formats and media, including some movie formats, may require additional plug-ins or helper applications.

4.2 Design Standards and Design Guidelines

Q.7 What is the definition of usability as per ISO 9241 standard ? Effective applications are both consistent within themselves and consistent with one another. Discuss that in context of Microsoft Office products.

[SPPU : May-18, End Sem, Marks 8]

Ans. :

- **Usability** : The effectiveness, efficiency and satisfaction with which specified users achieve specified goals in particular environments.
- **Effectiveness** : The accuracy and completeness with which specified users can achieve specified goals in particular environments.
- **Efficiency** : The resources expended in relation to the accuracy and completeness of goals achieved.
- **Satisfaction** : The comfort and acceptability of the work system to its users and other people affected by its use.
- The strength of a standard lies in its ability to force large communities to abide, the so-called authority.
- The authority of a standard (or a guideline, for that matter) can only be determined from its use in practice.
- Some software products become de facto standards long before any formal standards document is published, for example, the X windowing system).
- Usability test of Microsoft word :
 - Task 1 : Create New Document
 - Task 2 : Save Document under Custom Title
 - Task 3 : Choose Custom Font
 - Task 4 : Paragraph Formatting
 - Task 5 : Creating Table of Contents
 - Task 6 : Add Page Numbers
 - Task 7 : Upload/Format Image

Q.8 Explain basic categories of the Smith and Mosier guidelines. Also explain guidelines for data entry.

Ans. :

- Basic categories of the Smith and Mosier guidelines are as follows :

1. Data entry
2. Data display
3. Sequence control
4. User guidance
5. Data transmission
6. Data protection

- Smith and Mosier (1986) guidelines for data entry :

1. Consistency of data-entry transactions : Similar sequences of actions should be used under all conditions.
2. Minimal input actions by user : Greater productivity, fewer chances for error.
3. Minimal memory load on users : Should not be required to memorize lengthy lists of commands.
4. Compatibility of data entry with data display.
5. Flexibility for user control of data entry.

Q.9 Explain Shneiderman's eight golden rules of interface design.

Ans. :

- Shneiderman's eight Golden rules are as follows :

1. **Strive for consistency** : Consistent sequences of actions should be required in similar situations; identical terminology should be used in prompts, menus and help screens; and consistent commands should be employed throughout.
2. **Enable frequent users to use shortcuts** : As the frequency of use increases, so do the user's desires to reduce the number of interactions and to increase the pace of interaction. Abbreviations,

function keys, hidden commands, and macro facilities are very helpful to an expert user.

3. **Offer informative feedback** : For every operator action, there should be some system feedback. For frequent and minor actions, the response can be modest, while for infrequent and major actions, the response should be more substantial.

4. **Design dialog to yield closure** : Sequences of actions should be organized into groups with a beginning, middle and end. The informative feedback at the completion of a group of actions gives the operators the satisfaction of accomplishment, a sense of relief, the signal to drop contingency plans and options from their minds and an indication that the way is clear to prepare for the next group of actions.

5. **Offer simple error handling** : As much as possible, design the system so the user cannot make a serious error. If an error is made, the system should be able to detect the error and offer simple, comprehensible mechanisms for handling the error.

6. **Permit easy reversal of actions** : This feature relieves anxiety, since the user knows that errors can be undone; it thus encourages exploration of unfamiliar options. The units of reversibility may be a single action, a data entry or a complete group of actions.

7. **Support internal locus of control** : Experienced operators strongly desire the sense that they are in charge of the system and that the system responds to their actions. Design the system to make users the initiators of actions rather than the responders.

8. **Reduce short-term memory load** : The limitation of human information processing in short-term memory requires that displays be kept simple, multiple page displays be consolidated, window-motion frequency be reduced and sufficient training time be allotted for codes, mnemonics and sequences of actions.

4.3 What is Interaction Design ?

Q.10 What is design ? What is the golden rule of design ? Illustrate the process of interaction design.

Ans. [SPPU : May-18, Dec-17, End Sem, Marks 8]

Ans. :

- Design means achieving goals within constraint.
- Interaction design is the practice of designing interactive digital products, environments, systems and services.
- The golden rule of design are as follows :

1. Understand your materials : In the case of a physical design this is obvious. But for the chair with a steel frame and one with a wooden frame. They are very different : often the steel frames are tubular or thin L or H section steel.

- In contrast wooden chairs have thicker solid legs. If you made a wooden chair using the design for a metal one it would break; if you made the metal one in the design for the wooden one it would be too heavy to move.

- For Human-Computer Interaction the obvious materials are the human and the computer. That is we must :

- Understand computers : Limitations, capacities, tools, platforms
- Understand people : Psychological, social aspects, human error.

Interaction design process :

- Fig. Q.10.1 shows Interaction design process.
- **Requirements :** What is wanted ? The first stage is establishing what exactly is needed. There are a number of techniques used for this in HCI : interviewing people, videotaping them, looking at the documents and objects that they work with, observing them directly.

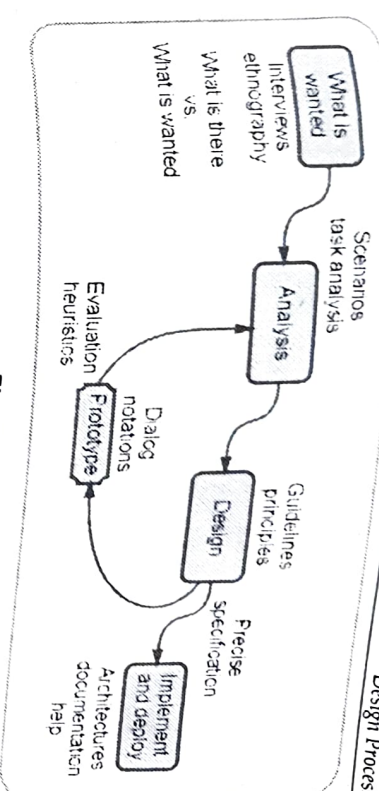


Fig. Q.10.1

- **Analysis :** The results of observation and interview need to be ordered in some way to bring out key issues and communicate with later stages of design.

- **Design :** There are numerous rules, guidelines and design principles that can be used to help Iteration and prototyping. Humans are complex and we cannot expect to get designs right first time. We therefore need to evaluate a design to see how well it is working and where there can be improvements.

- **Iteration and prototyping :** Humans are complex and we cannot expect to get designs right first time. We therefore need to evaluate a design to see how well it is working and where there can be improvements.

- **Implementation and deployment :** This will involve writing code, perhaps making hardware, writing documentation and manuals - everything that goes into a real system that can be given to others.

Q.11 Explain different stages in the design.

Ans. :

- **Concept design :** Every possible approaches are sketched out. The validity of each sketches is verified following the usability requirements and the goals agreed. At the end, the best approach is selected.

- **Interaction design** : In this step, the structure of the UI must be set by naming interaction flows. Method : e.g. Affinity diagramming, each action can be written on a Post-It note and organised in clusters. The Post-It notes are then rearranged to simplify user tasks.
- **Screen design** : Creating rough designs of the screens' structure. These layouts are linked together and usability test is performed with a user.
- **Testing** : A user is asked to follow a realistic scenario/tasks on the sketched out prototype.

Q.12 Write short note on process on interaction design with respect to following points : i) Basic activities ii) Characteristics

[SPPU : May-19, End Sem, Marks 8]

Ans. : i) **Basic activities**

- Interaction design involves four basic activities :
 1. Identifying needs and establishing requirements.
 2. Developing alternative designs that meet those requirements.
 3. Building interactive versions of the designs so that they can be communicated and assessed.
 4. Evaluating what is being built throughout the process.
- These are permeated with three principles :
 1. Involve users early in the design process and evaluation of the artifact
 2. Define quantifiable and measurable usability criteria
 3. Iteration is inevitable
- Key characteristics of the interaction design process are explicit incorporation of user involvement, iteration and specific usability criteria.
- Before you can begin to establish requirements, you must understand who the users are and what their goals are in using the device.

- Looking at others' designs provides useful inspiration and encourages designers to consider alternative design solutions, which is key to effective design.
- Usability criteria, technical feasibility and users' feedback on prototypes can all be used to choose among alternatives
- Prototyping is a useful technique for facilitating user feedback on designs at all stages.

ii) **Characteristics**

1. It changes situations by shaping and deploying artifacts.
2. It explores possible futures.
3. It frames the problem in parallel with possible solutions.
4. It requires thinking through "sketching" and other "tangible representations"
5. It addresses instrumental, technical, aesthetic and ethical aspects throughout the process.

Q.13 Write a scenario for music player design ?

[SPPU : May-19, End Sem, Marks 8]

Ans. :

- Interaction design is about creating interventions in often complex situations using technology of many kinds including PC software, the web and physical devices.
- Scenarios are rich design stories, which can be used and reused throughout design :
 1. They help us see what users will want to do
 2. They give a step-by-step walkthrough of users' interactions: including what they see, do and are thinking.
- The main work of music player is to read the audio file: the file decoding and information display. Therefore, the overall design of an music player can be divided into the following three processing stages :
 - a) Accessing audio files, mainly including the transmission mode and the choice of storage media.

- b) Decoding audio files, the main decoding scheme is the choice of soft or hard to decode.
- c) Audio file information display, including the selection of the display, mainly to add more features and gorgeous interface to increase the user experience.
- The design includes physical artifacts the scenarios can be used as a script to act out potential patterns of use.
- Scenarios can be used to : communicate with others - other designers, clients or users. It is easy to misunderstand each other whilst discussing abstract ideas. Concrete examples of use are far easier to share.
- Validate other models A detailed scenario can be 'played' against various more formal representations such as task models or dialog and navigation models.
- Express dynamics Individual screen shots and pictures give you a sense of what a system would look like, but not how it behaves.

4.4 User Focus, Scenarios

Q.14 How to get to know the system users ? Explain various methods adopted in user - centered design. What are the people directly or indirectly affected by a student registration system ?

Ans. : [SPPU : May-18, End Sem, Marks 8]

- User Centred Design (UCD) is an approach to interactive system development that focuses specifically on making systems or applications easy to use.
- The purpose of User Centred Design (UCD) is to involve end users in the development process of the product or system in a way that the prototypes and designs and finally the products or systems would meet the needs and requirements of the users as well as possible.

- Usability can be defined as the measure of the quality of a user's experience when interacting with a product or service
 - Usability is recognized as one of the most important characteristics of systems and products. Usable systems are easy to learn, efficient to use, not error prone and satisfactory in use.
 - 1. With close user involvement, products are more likely to meet users' expectations and requirements. This leads to increased sales and lower costs incurred by customer services.
 - 2. Systems designers tailor products for people in specific contexts and with specific tasks, thereby reducing the chances of situations with a high risk of human error arising. UCD leads to safer products.
 - 3. Putting designers in close contact with users means a deeper sense of empathy emerges. This is essential in creating ethical designs that respect privacy and the quality of life.
 - 4. By focusing on all users of a product, designers can recognize the diversity of cultures and human values through UCD - a step in the right direction towards creating sustainable businesses.
 - People directly or indirectly affected by a student registration system are university, college, teacher, student, parents, timetable scheduler etc.
- Q.15 A scenario is an idealized but detailed description of a specific instance of human - computer interaction (HCI). Scenarios specify how users carry out their tasks in specified context. Write scenarios for purchasing an airline ticket.**
- Note - Generate scenarios to cover a wide range of situations, not just the most common ones. Include problem situations that will test the system concept, not just straightforward scenarios.**

Ans. : [SPPU : Dec-17, May-18, End Sem, Marks 8]

- Scenarios are the core of any usability test as you have to correctly communicate your participants, what you want them to do during the test. Well crafted task scenario helps you focus on designing around the real needs of the user and removes artificiality from the test.

- As you can see, above scenario makes the task much more realistic as we have provided user only a scenario which requires users to complete 3 different tasks-

1. Signup login on the app
2. Search for flight as per the schedule
3. Book the flight

- A good scenario -

1. Short but enough information to perform the task
2. Use user's language, not the product's
3. Simple and clear
4. Should address your tasks and concerns

1. Short but enough information to perform the task :

- Time needed to read and understand the task has to be minimized as much as possible. Having a long written task scenarios may require users to spend undue time in reading and understanding what they have to do during the test which may indirectly influence the overall time and effort to complete the task.

- You have to find a balance in keeping the scenario short and relieving enough information to perform the task. It is also suggested to communicate the task with users as the way you talk and not sound very scientific.

2. Use user's language, not the product's :

- Main aim of conducting usability test is to understand how a user will use it in their real environment without getting any support from external audience.
- So, providing users with the enough detail is important but it should be in the language that user can relate to and not the one which is used in product.
- For example, You may have Icons, menu options or labeled button, in your UI. Concern could be to see if users choose right Icons, menu options or labeled button to complete a task or not.

3. Simple and Clear :

- You will get desired result if and only if your task scenario is clear to your users that means they have no ambiguity in understanding what you want them to do.

4. Should address your tasks and concerns :

- Every scenario we create has to address one or more tasks and each task should be intended to address one or more concerns we have with the app/ website.
- It's must to have scenarios that are aligned with your business goals, for example, if your concern is to improve sales in your e-commerce portal, you would probably be interested in knowing whether it's easy for your customers to find and purchase a given product or not.

Q.16 Write short note on user focus.

Ans. :

- User focus is a never ending process because there is so much to know and because the users keep changing.
- An interactive system designer should consider the human factors that characterize users.
- User characteristics vary with age, gender, physical and cognitive abilities, personality, education, cultural or ethnic background and goals.
- An interactive system designer should recognize this diversity. Systems used by several communities of users.
- Designer faces real challenge to cater to the need of each community. Designers must characterize users and situations as precisely and completely as possible.
- Over time many people are affected directly or indirectly by a system and these people are called stakeholders.
- Tracing the tenuous links between people could go on forever and user need to draw boundaries as to whom you should consider. This depends very much on the nature of the systems being designed.

- When designing a system it is easy to design it as if you were the main user : you assume your own interests and abilities.
- People may also be able to tell you about how things really happen, not just how the organization says they should happen. To encourage users to tell you this, you will need to win their trust, since often the actual practices run counter to corporate policy.
- A professional in any field is very practiced and can do things in the domain. An academic in the same field may not be able to do things, but she knows about the things in the domain. These are different kinds of knowledge and skill.
- Sometimes people know both, but not necessarily so. The best sports trainers may not be the best athletes, the best painters may not be the best art critics.
- Because of this it is important to watch what people do as well as hear what they say. This may involve sitting and taking notes of how they spend a day, watching particular activities, using a video camera or tape recorder.
- It can be done in an informal manner or using developed methods such as ethnography or contextual inquiry.
- Another way to find out what people are doing is to look at the artifacts they are using and creating. Look at a typical desk in an office. There are papers, letters, files, perhaps a stapler, a computer, sticky notes.
- One method that has been quite successful in helping design teams produce user focused designs is the persona. A persona is a rich picture of an imaginary person who represents your core user group.

4.5 Navigation Design and Screen Design

Q.17 What is navigation design ? Explain its local structure.

Ans. :

Navigation Design : Imagine yourself using a word processor. You interact at several levels :

- **Widgets** help you know how to use them for a particular selection or action.
- **Screens or windows** - To understand the logical grouping of buttons.
- **Navigation within the application** - To understand where you are in the interaction.
- **Environment** - You swap between applications, perhaps cut and paste.
- In the web we have less control of how people enter a site and on a physical device we have the same layout of buttons and displays no matter what the internal state (although we may treat them differently). Just in case you haven't already got the idea, the place to start when considering the structure of an application is to think about actual use :

- 1) Who is going to use the application ?
 - 2) How do they think about it ?
 - 3) What will they do with it ?
- This can then drive the second task - thinking about structure.
 - We will consider two main kinds of issue :

- 1) Local structure : Looking from one screen or page out
- 2) Global structure : Structure of site, movement between screens.

Local structure

- Much of interaction involves goal-seeking behavior. In an ideal world if users had perfect knowledge of what they wanted and how the system worked they could simply take the shortest path to what they want.
- At each point in the interaction they can make some assessment of whether they are getting closer to their (often partially formed) goal.

- To do this goal seeking, each state of the system or each screen needs to give the user enough knowledge of what to do to get closer to their goal. To get you started, here are four things to look for when looking at a single web page, screen or state of a device.
 1. Knowing where you are
 2. Knowing what you can do
 3. Knowing where you are going - or what will happen
 4. Knowing where you've been - or what you've done
- The screen, web page or device displays should make clear *where you are* in terms of the interaction or state of the system. Some websites show 'bread crumbs' at the top of the screen, the path of titles showing where the page is in the site. Trade-off between appearance and ease of use may mean that this is the right thing to do, but you should take care before confusing the user needlessly.
- You need to know *where you are going* when you click a button or *what will happen*. It is better if users do not have to use this 'try it and see' interaction. Icons are typically not self-explanatory and should always be accompanied by labels or at the very least tooltips or some similar technique.
- Special care has to be taken if the same command or button press means something different in different contexts. system needs to give some *feedback* to say what has happened.

Q.18 If the user has perfect knowledge of what they wanted (goal) and how the system worked (task) interaction becomes effective and satisfying. In designing navigation for a website page each screen needs to give the user enough knowledge of what to do to get closer to their goal. Discuss four important questions that drive a webpage navigation design for achieving the above motto.

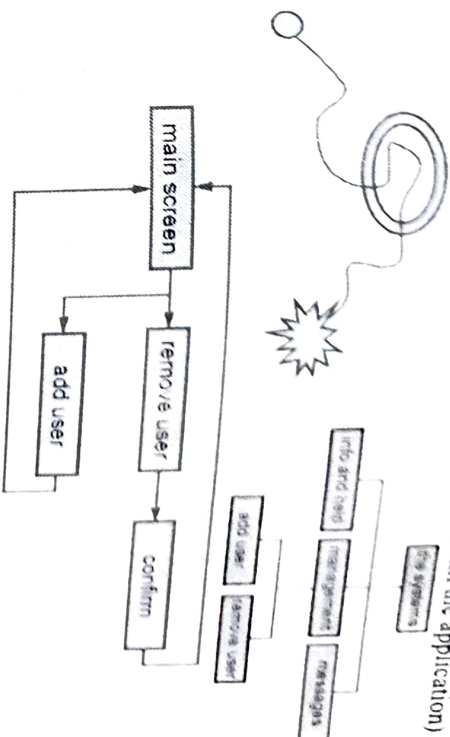
Ans. [SPPU : Dec.-17, End sem, Marks 8]

Ans. : Navigation Design

Local structure - Single screen (From one screen looking out)

What are the issues or concerns of local structure ?

- Q.1 knowing where you are
 - Q.2 knowing what you can do
 - Q.3 Knowing where you are going or what will happen
 - Q.4 knowing where you've been or what you've done
- Global structure - Whole site (between screens within the application)



Q.19 Write short note on wider still.

Ans. :

- Each sits amongst other devices and applications and this in turn has to be reflected within our design.
- This has several implications :

- 1. Style issues :** We should normally conform to platform standards, such as positions for menus on a PC application, to ensure consistency between applications. For example, on our proposed personal movie player we should make use of standard fast-forward, play and pause icons.
- 2. Functional issues :** On a PC application we need to be able to interact with files, read standard formats and be able to handle cut and paste.
- 3. Navigation issues :** We may need to support linkages between applications, for example allowing the embedding of data from one application in another or, in a mail system, being able to

double click an attachment icon and have the right application launched for the attachment.

Q.20 While design screen, which of the appropriate appearance are considered ?

Ans. : Presenting information :

- The way of presenting information on screen depends on the kind of information: text, numbers, maps, tables.
- Technology available to present it : character display, line drawing, graphics, virtual reality; and, most important of all, on the purpose for which it is being used.

Aesthetics and utility

- Remember that a pretty interface is not necessarily a good interface. Ideally, as with any well-designed item, an interface should be aesthetically pleasing.
- The conflict between aesthetics and utility can also be seen in many 'well designed' posters and multimedia systems.
- In particular, the backdrop behind text must have low contrast in order to leave the text readable; this is often not the case and graphic designers may include excessively complex and strong backgrounds because they look good. The results are impressive, perhaps even award winning, but completely unusable.

Making a mess of it : Color and 3D

- The increasing use of 3D effects in interfaces has posed a whole new set of problems for text and numerical information.

Localization / Internationalization

- The process of making software suitable for different languages and cultures is called localization or internationalization.

Q.21 What is screen design ?

Ans. :

- Screen design refers to the graphic design and layout of user interfaces on displays.

- The basic principle at the screen level reflect those in other areas of interaction design.:

1. Ask 2. Think 3. Design

- It is a sub-area of user interface design but is limited to monitors and displays.

- In screen design, the focus is on maximizing usability and user experience by making user interaction as simple and efficient as possible.

Q.22 Explain the following with respect to screen design :

a. Layout b. Grouping structure c. White space

Ans. :

a. Layout :

- Layout is the arrangement of items on the screen. Like items are grouped into areas. Fig. Q.22.1 shows grouping related items in an order screen.

Billing details:		Delivery details:	
Name:		Name:	
Address:		Address:	
Order No:		Deliver date & time:	
Order Details:			
Name of Item	Quantity	Price	Cost
Book	2	210/-	420/-

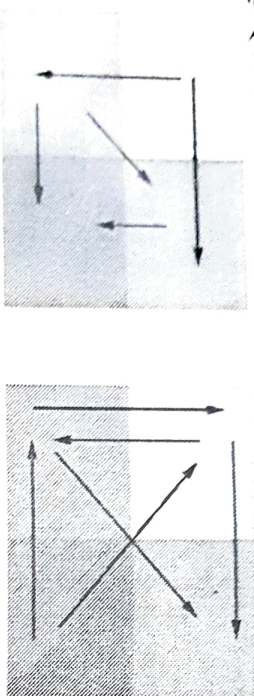
Fig. Q.22.1

- Areas can be further subdivided and each area is self-contained. Areas should have a natural intuitive flow.
- Users from western nations tend to read from left to right and top to bottom. Users from other regions may have different flows.
- Number of visual tools are available to help us suggest to the user appropriate ways to read and interact with a screen or device.

b. Grouping and structure :

- Logical things are physically grouped together. This may involve multiple levels of structure. Above figure shows a potential design for an ordering screen.
- Group data by the natural sequence of use.
- Flow of control : how users progress through a screen when doing their work.

- Fig. Q.22.2 shows flow of control.

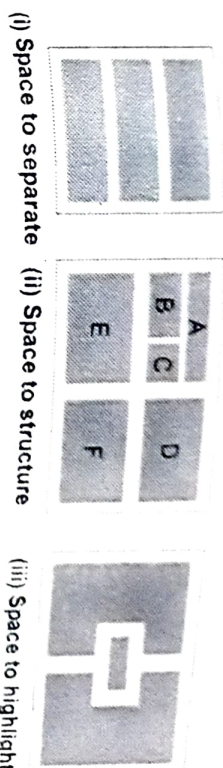


Which flow of control is correct?

Fig. Q.22.2

- Flow of control means that the focus of activity moves across a screen or page while the user performs a certain task.
- Flow of control is important for (1) efficiency in performing a task (2) transparency and understandability of a screen or page.
- For Western cultures the natural flow is from left to right and from top to bottom.
- c. White space :**
- One way to make an element within an item stand out in a list UI design is to simply leave it alone literally.
- White space is one of the best ways to isolate the important elements in a list item. Doing this draws the user's attention more easily. Essential role in screen layout. Use margins to draw eye around design.

- White space is the area between design elements. It is also the space within individual design elements, including the space between typography glyphs (readable characters).
- Despite its name, white space does not need to be white. It can be any color, texture, pattern or even a background image.
- Fig. Q.22.3 shows use of white space.

**Fig. Q.22.3**

- Try to integrate figure and ground. Object should be scaled proportionally to its background.
- Don't crowd controls together because crowding creates spatial tension and inhibits scanning.
- The advantages of using whitespace :
 1. Increased content legibility
 2. More interaction
 3. Ability to highlight call to actions
 4. Act as a separator.

4.6 Prototyping Techniques and Wire-Framing

Q.23 What is a prototype ? Explain different types of rapid prototyping techniques.

[SPPU : Dec.-17, May-18, End Sem, Marks 8]

Ans. :

- A **prototype** is an example that serves as a basis for future models. Prototyping gives designers an opportunity to research new

- alternatives and test the existing design to confirm a product's functionality prior to production.
- Prototyping is an example of what is known as a hill-climbing approach.
- Rapid prototyping** is an instructional design approach that combines the design, developmental and evaluation phases. It is a non-linear approach that produces a sample working model that is a scaled-down representative version of the whole course.
- Three main approach of prototyping are throw-away, incremental and evolutionary.

Types of rapid prototyping :

- Storyboard, paper prototype, wireframes and mock up review form are the types of rapid prototyping.
- 1. Storyboard :** It is a graphical depiction of the outward appearance of the intended system, without any accompanying system functionality. Storyboards do not require much in terms of computing power to construct.
- Storyboarding allows you to check your design is on target with expectations before investing time in developing.
- Modern graphical drawing packages now make it possible to create storyboards with the aid of a computer instead of by hand.
- 2. Paper prototype :** Easy and fast to do. It helps you think of specifics. Usually good as a first round prototype. It can still do usability testing, even with paper.
- 3. Wireframe :** Wireframes are the first stage of the design process and help us to understand the key user journey's, information structuring, modes of interaction and functionality.
- Wireframes or 'page schematic' are a basic outline or skeleton of your key website pages, drawn to show the elements of a page, their relationships, position and their relative importance.

- They indicate the information types present, navigation, signposting, branding and content areas. They are black and white schematics presented either in PowerPoint or as a clickable web prototype.
- 4. Mock-up** is a scale or full-size model of a design or device, used for design evaluation, promotion. A mockup is a prototype if it provides at least part of the functionality of a system and enables testing of a design. Mock-ups are used by designers mainly to acquire feedback from users.

Q.24 Write short note on wire-framing.

Ans. :

- The interaction modeling and interface options can be put together concretely using the so-called wire-framing process.
- Wire-framing originated from making rough specifications for website page design and resembles scenarios or storyboards.
- Usually, wire-frames look like page schematics or screen blueprints, which serve as a visual guide that represents the skeletal framework of a website or interface. It depicts the page layout or arrangement of the UI objects and how they respond to each other.
- Wireframes can be pencil drawings or sketches on a whiteboard or they can be produced by means of a broad array of free or commercial software applications.

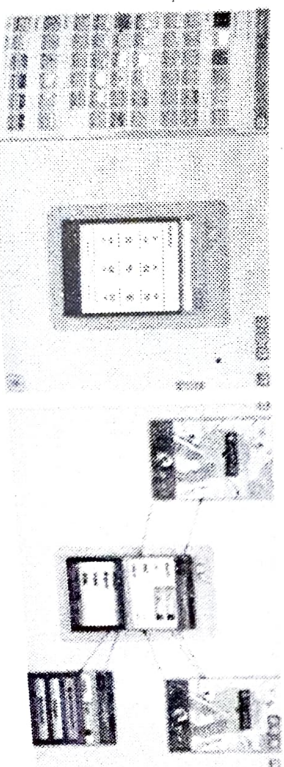


Fig. Q.24.1 : An example of a wire-framing tool

- Fig. Q.24.1 shows such a wire-framing tool. Wireframes produced by these tools can be simulated to show interface behavior and

depending on the tools, the interface logic can be exported for actual code implementation (but usually not).

- Note that there are tools that allow the user to visually specify UI elements and their configuration and then automatically generate code. Regardless of which type of tool is used, it is important that the design and implementation stages be separated.

- Through wire-framing, the developer can specify and flesh out the kinds of information displayed, the range of functions available and their priorities, alternative sand interaction flow.

Q.25 Explain problems with prototyping.

Ans. :

- The main problem with prototyping is that no matter how cheaply designers try to make their prototypes and how committed they are to throwing them away, they still have to make design decisions about how to present their prototypes.
- These decisions can then become ingrained into the final product. There is a factor to consider called 'design inertia' which describes how designers, once having made a design decision, are rather reluctant to relinquish that decision, admit it is wrong and change it.
- This is another reason why good designers put off big decisions as much as possible, knowing that if they make a small mistake early on it is easier and cheaper to rectify than a big early mistake.
- Rapid prototyping is about exposing bad design decisions as soon as possible after they have been made.
- Guidelines and usability engineering are more about trying to get the designer not to make mistakes in the first place.
- Design inertia is therefore much more prevalent in rapid prototyping development than it is in other user centred design processes. Because prototypes do not get thrown away in iterative design processes then design inertia is even more prevalent.

- Furthermore in spotting usability problems by user testing, the designer knows that there is a problem, but not necessarily what causes that problem or how to fix it. Rapid prototyping identifies symptoms, not illnesses and not cures.

Q.26 Explain Hill climbing approach with prototyping?

 [SPPU : Dec.-18, End sem, Marks 8]

Ans. :

- Hill climbing is a mathematical optimization heuristic method used for solving computationally challenging problems that have multiple solutions.
- It is an iterative method belonging to the local search family which starts with a random solution and then iteratively improves that solution one element at a time until it arrives at a more or less optimized solution.
- Prototyping is an example of what is known as a hill-climbing approach.
- A prototype is an early sample, model or release of a product built to test a concept or process or to act as a thing to be replicated or learned from.
- It is a term used in a variety of contexts, including semantics, design, electronics and software programming.
- A prototype is designed to test and try a new design to enhance precision by system analysts and users.
- When you're creating a prototyping strategy, it's important to think about the cost of change over time.
- For physical products like a car, a toaster, the cost of making changes rises dramatically over time throughout the design process and even more significantly upon release.
- With desktop software that gets distributed on, say, a CDROM, the cost rises aren't quite so dramatic, but it's still pretty significant.

- harder to make changes as you go throughout the design process and much more difficult once you've shipped it out to consumers.
- Web sites and other forms of software as a service, make it much easier to make changes over time.
 - But the costs and difficulty of making changes is still increasing for a number of reasons. Fig. Q.26.1 shows hill climbing approach.

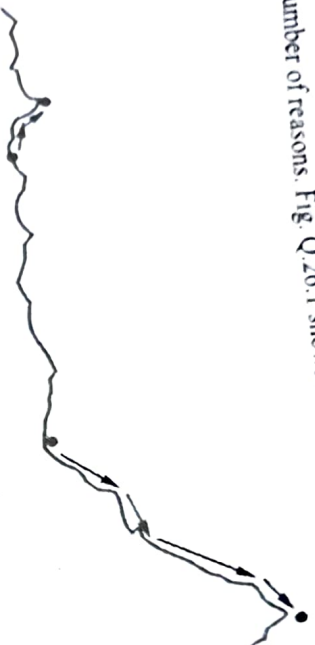


Fig. Q.26.1 : Hill climbing approach

Problems :

1. May hit a local maximum, not global one
 2. Need good starting point, but how to find?
- Standard mechanism for finding best starting point : use experienced designers, try different designs.
 - According to Norman, current UCD processes achieve good design. For inspirational great design, new ways to find starting points may be needed.

4.7 Understanding the UI Layer and its Execution Framework

Q.27 Explain user interface layer with its execution framework.

Ans. :

- The UI layer refers to a set of software that operates above the core operating system (and underneath the application). It encapsulates and exposes system functions for :
 - Fluid input and output (I/O)

- Facilitation of development of I/O functionalities (in the form of an application programming interface/library [API] or toolkit)
- Run-time management of graphical applications and UI elements often manifested as windows or graphical user interface (GUI) elements (in the form of separate application often called the window manager)
- Since most interfaces are graphical, the UI layer uses a two- or three-dimensional (2-D or 3-D) graphical system based on which GUI elements are implemented.
- Thus, to summarize, the UI layer is largely composed of (a) An API for creating and managing the user interface elements (e.g., windows, buttons, menus) and (b) A window manager to allow users to operate and manage the applications through its own user interfaces.
- As shown in Fig. Q.27.1 the user interacts with the window/GUI-based applications using various input and output devices. At the same time, aside from the general applications, the user interacts with the computer and manages multiple application windows/tasks.
- The window manager is regarded as both an application and API. User applications are created using the APIs that represent abstracted I/O-related functionalities of the UI layer, such as those for window managing (resizing, iconifying, dragging, copy and paste, etc.), GUI elements and widgets (windows, menus, buttons, etc.) and basic windowing (creating/destroying window, deactivating window, etc.).

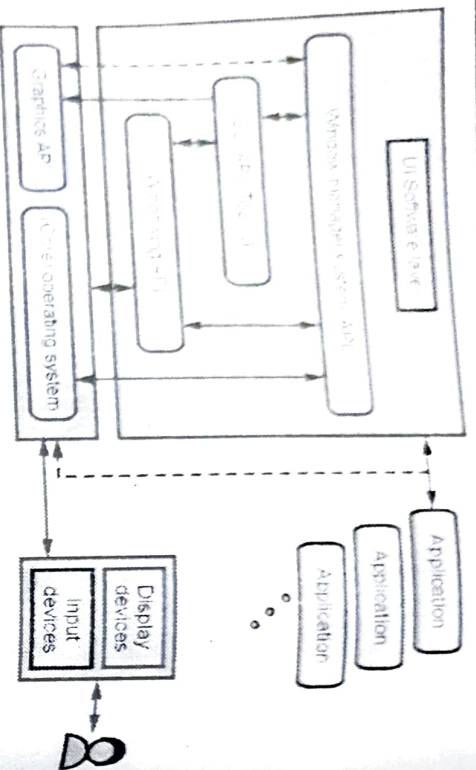


Fig. Q.27.1 : User interface software layer for a window-based multitasking UI

4.8 Model-View-Controller (MVC) Framework

Q.28 What is MVC framework ? Explain its components.

Ans. :

- The Model-View-Controller (MVC) is an architectural pattern that separates an application into three main logical components: the model, the view and the controller.
- Each of these components are built to handle specific development aspects of an application. MVC is one of the most frequently used industry-standard web development framework to create scalable and extensible projects.
- A standard design pattern of graphical user interfaces that is used at many levels, including overall application design and individual visual components.

MVC Components :

- Interface architecture is decomposed into three parts :
 1. **Model** : manages data and its manipulation
 2. **View** : manages the presentation of the data
 3. **Controller** : manages user interaction

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- Fig. Q.28.1 shows MVC architecture.
- Model is decoupled by View, View and Controller know all about model. In practice, View and Controller are often coupled. View knows to send events to Controller.

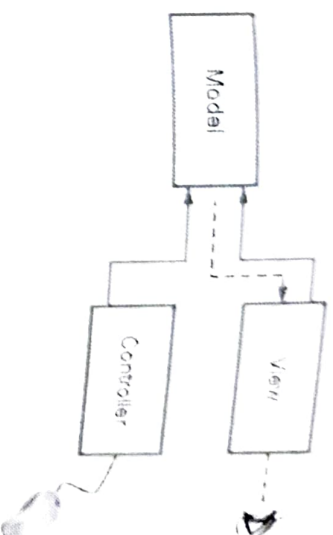


Fig. Q.28.1

- Controller knows about view modern widget toolkits use MVC throughout. Simple widgets (e.g., buttons, checkboxes, scrollbars) usually contain a default model within themselves.
- **Model** : Model component corresponds to all the data-related logic that the user works with. This can represent either the data that is being transferred between the View and Controller components or any other business logic-related data.
- **View** : The View component is used for all the UI logic of the application. For example, the Customer view will include all the UI components such as text boxes, dropdowns, etc. that the final user interacts with.
- **Controller** : Controllers act as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the model component and interact with the Views to render the final output.
- For example, the customer controller will handle all the interactions and inputs from the customer view and update the database using the customer model. The same controller will be used to view the customer data.

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